

Problem name: Implement FCFS Algorithm.

Source code:

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    int n;
    cout << "Enter total number of processes: ";
    cin >> n;

    vector<int> burst(n), waiting(n, 0), turnaround(n), completion(n);

    for (int i = 0; i < n; i++) {
        cout << "Enter burst time of P" << i + 1 << ": ";
        cin >> burst[i];
    }

    for (int i = 1; i < n; i++) {
        waiting[i] = waiting[i - 1] + burst[i - 1];
    }

    for (int i = 0; i < n; i++) {
        completion[i] = waiting[i] + burst[i];
        turnaround[i] = completion[i];
    }

    float sumWT = 0;
    for (int i = 0; i < n; i++) {
        sumWT += waiting[i];
    }

    float avgWT = sumWT / n;

    cout << "\nProcess \t BT \t WT \t TAT \t CT\n";
    for (int i = 0; i < n; i++) {
        cout << "P" << i + 1 << "    \t"
            << burst[i] << "    \t"
            << waiting[i] << "    \t"
            << turnaround[i] << " \t "
            << completion[i] << endl;
    }

    cout << "\nAverage Waiting Time = " << avgWT << endl;

    return 0;
}
```

Sample input:

```
Enter total number of processes: 5
Enter burst time of P1: 12
Enter burst time of P2: 3
Enter burst time of P3: 12
Enter burst time of P4: 43
Enter burst time of P5: 3
```

Sample output:

Process	BT	WT	TAT	CT
P1	12	0	12	12
P2	3	12	15	15
P3	12	15	27	27
P4	43	27	70	70
P5	3	70	73	73

Average Waiting Time = 24.8